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Course Number 074742056201

Thermodynamical Cycle Processes



Questions and Exercises about Energy Markets

- ♦ Questions: Appendix A.1, page 133
- ♦ Exercises: Appendix A.2.1, page 159

Introduction Thermal Power Plant

- ◆ Several types of fuel
 - Coal
 - Gas
 - Oil
- ◆ Heat a boiler
- ◆ Working material
 - Steam
 - Gas
- ◆ Drives Generator
- ◆ Produces electricity

Theoretical principles – Convert Heat To Work

- ♦ Conversion done via cycle processes (dt. Kreisprozesse)
- ♦ Working material heated up to high temperatures
- ♦ Working material is
 - carrying out the actual mechanical work
 - Give off low temperature heat
- ♦ Main property of an cycle process:

After several state changes
the final state is identical to the initial state

Carnot Cycle

- ◆ Nicolas Leonard Sadi Carnot (1796 – 1832)
- ◆ French Engineer
- ◆ Founder of the science of thermodynamics
- ◆ First to recognize the relationship between work and heat
- ◆ Carnot cycle describes the state changes of an working material
 - E.g. ideal gas within a cylinder



Carnot Cycle

- ◆ 1-2
 - Heat is supplied to the ideal gas
 - Isothermal expansion
- ◆ 2-3
 - Work is dispensed
 - Isentropic expansion
- ◆ 3-4
 - Heat is dispensed
 - Isothermal compression
- ◆ 4-1
 - Work is supplied
 - isentropic compression

First law of thermodynamics

- ♦ Energy supplied to a system is equal the change in stored energy is equal to the internal energy u
- ♦ Because in a circle process the initial and end state are the same, *internal energy u is not changing*

dq : supplied heat

dw : work performed

$$Q_{du} = \oint dq - \oint dw = 0$$

This equation is valid for all integration paths, thus:

$$du = dq - dw = dq - pdv$$

Carnot Cycle

- ◆ States of working material can be described

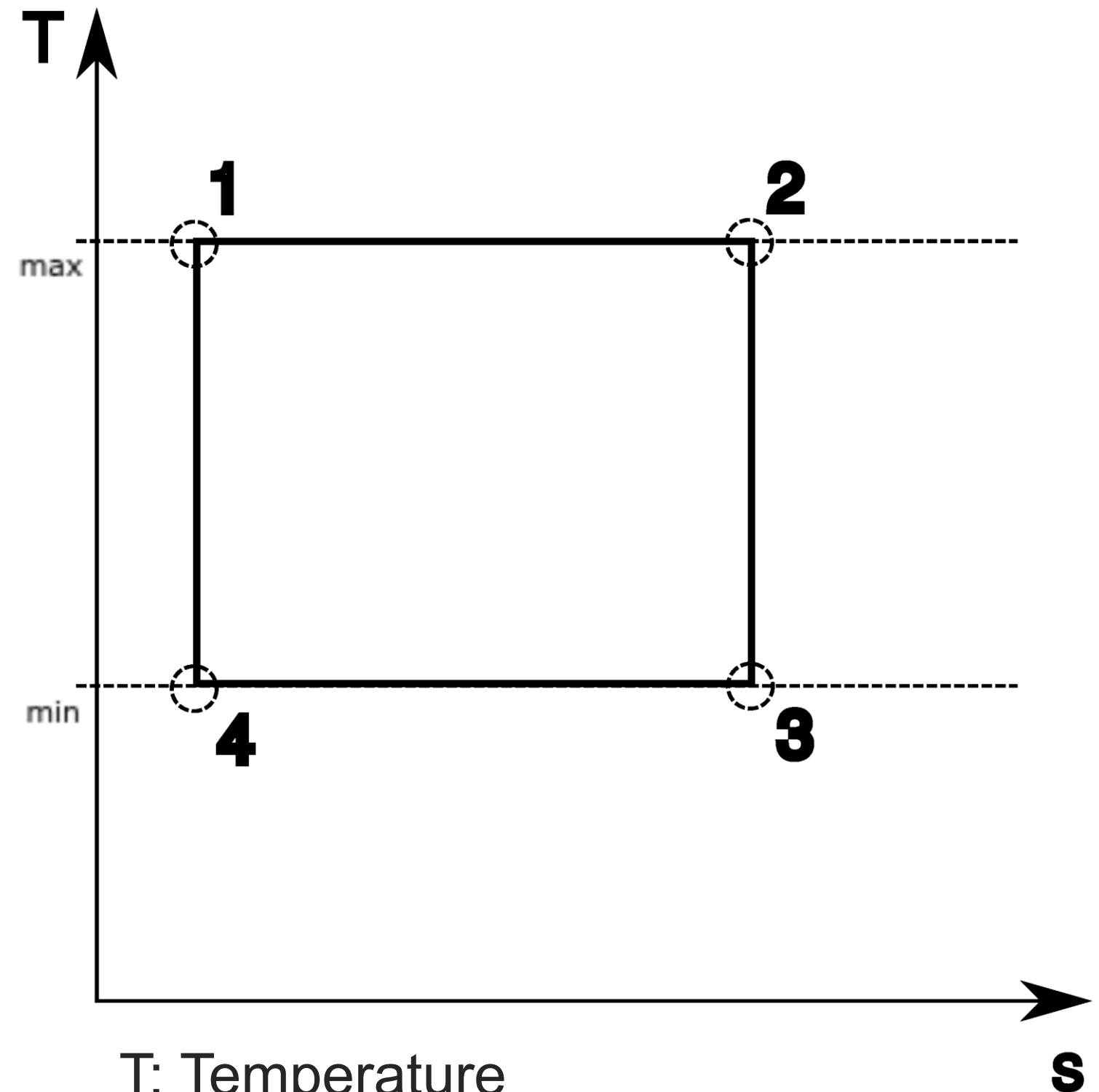
- p-v diagram
- T-s or h-s diagram

- ◆ 1-2

- Isothermal expansion when heat is supplied $q_{12} = q_{\text{Sup}}$
- w_{12} is dispensed

- ◆ 3-4

- Isothermal compression when heat is dispensed $q_{34} = q_{\text{dis}}$
- work w_{34} is supplied



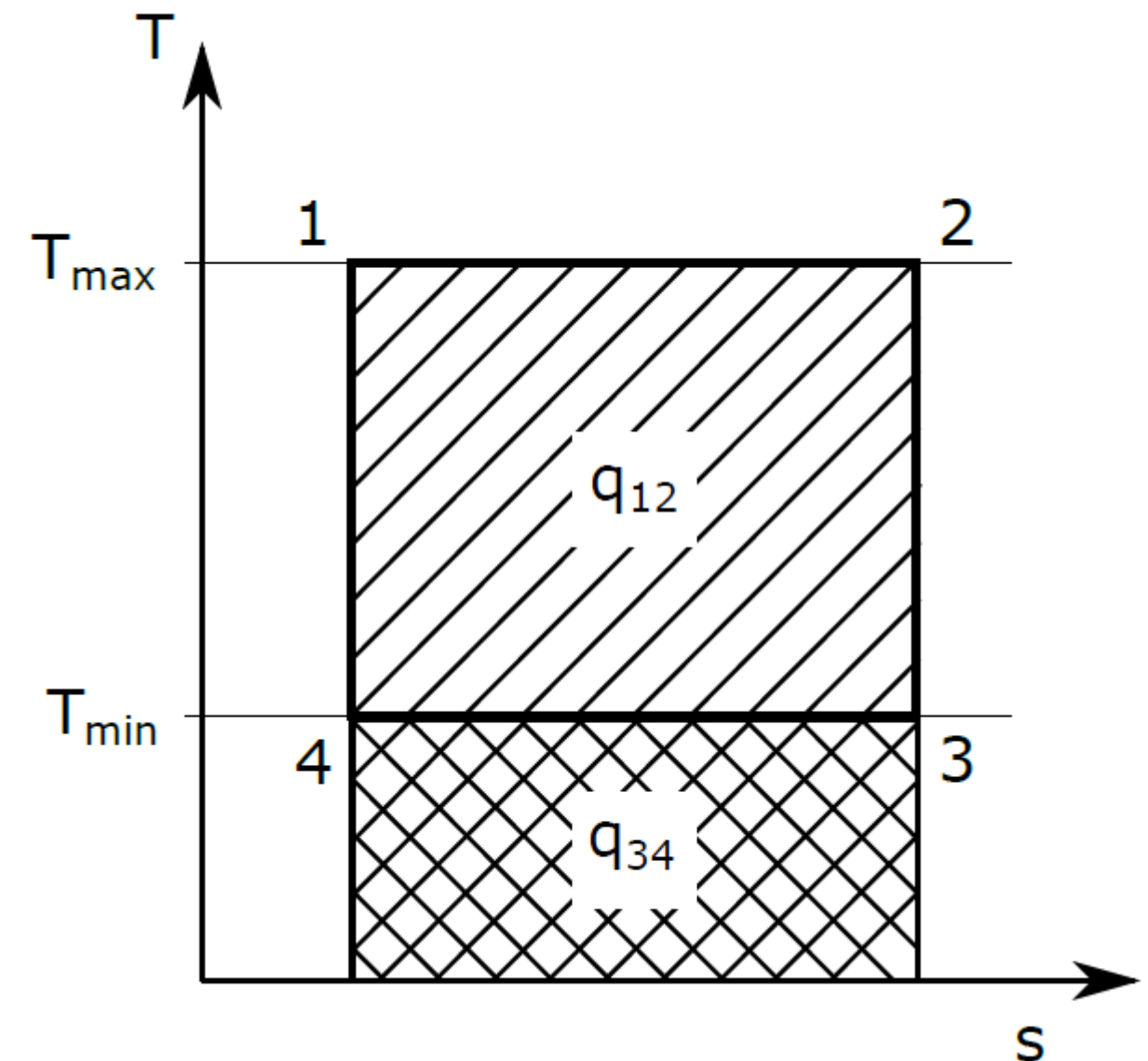
T: Temperature

s: Specific entropy

Heat

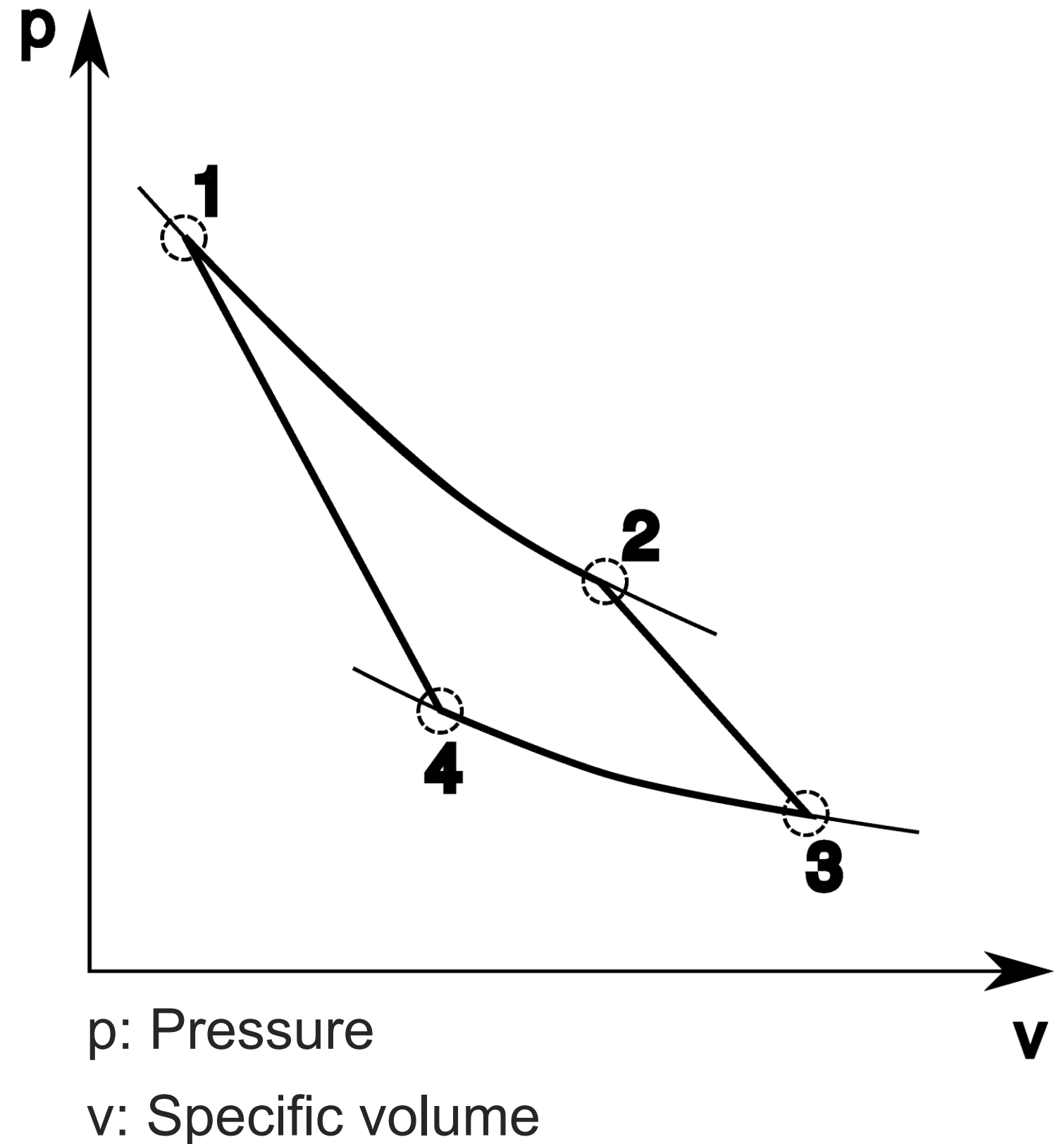
- ♦ Supplied and dispensed heat are equal the area within the T-s diagram
- ♦ Entropy is defined as $dq = T ds$
- ♦ Sum of dispensed and supplied heat can be calculated:

$$q_{sum} = \oint dq = \oint T ds$$



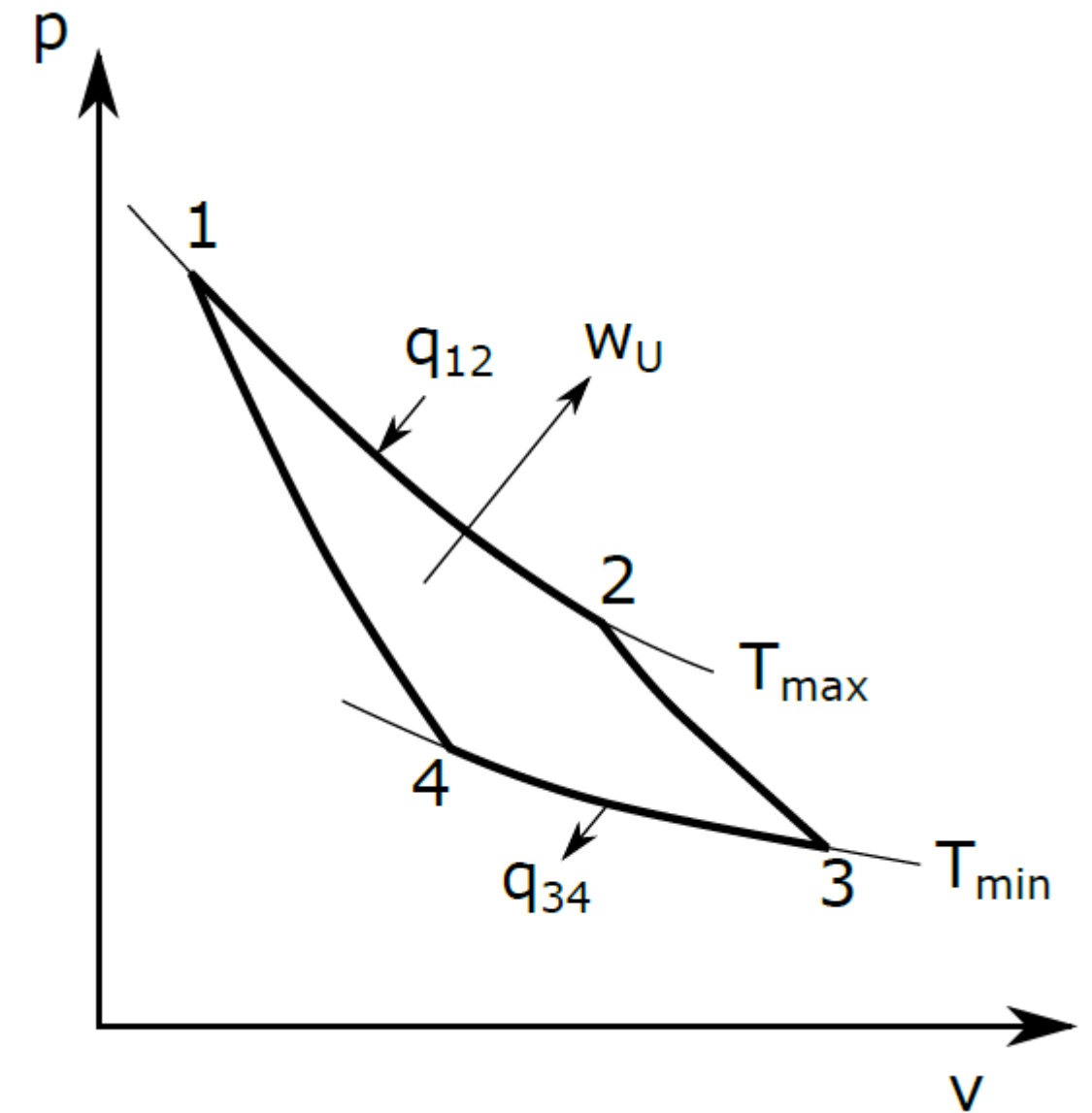
Carnot Cycle

- ♦ States of working material can be described
 - p-v diagram
 - T-s or h-s diagram
- ♦ 2-3
 - Isentropic expansion when work w_{23} is dispensed
- ♦ 4-1
 - Isentropic compression when work w_{41} is supplied



Dispensed work

- ◆ State changes of the cycle process clock wise for prime movers
- ◆ Prime mover (dt. Antriebsmaschine)
 - Thermal energy gets converted to mechanical energy



Useful Work

- ♦ Useful work w_u which can be extracted from the process:

$$w_U = \oint dw = \oint dq = q_{sup} - q_{dis}$$

- ♦ Useful work is equal the difference between the supplied and dispensed heat

Reversible Carnot Process

- ♦ Carnot process shows that in order to get mechanical work from heat a cycle process is used
 - heat is supplied on high temperature level
 - heat must be withdrawn on a low temperature level
- ♦ Because the circle integral of $ds = 0$ we can write:

$$\oint ds = \oint \frac{dq}{T} = \frac{q_{12}}{T_1} + \frac{q_{34}}{T_4} = \frac{q_{sup}}{T_{max}} - \frac{q_{dis}}{T_{min}}$$

Reversible Carnot Process

... continued

- ♦ Based on the previous integral we can say

$$|q_{ab}| = \frac{T_{min}}{T_{max}} q_{sup}$$

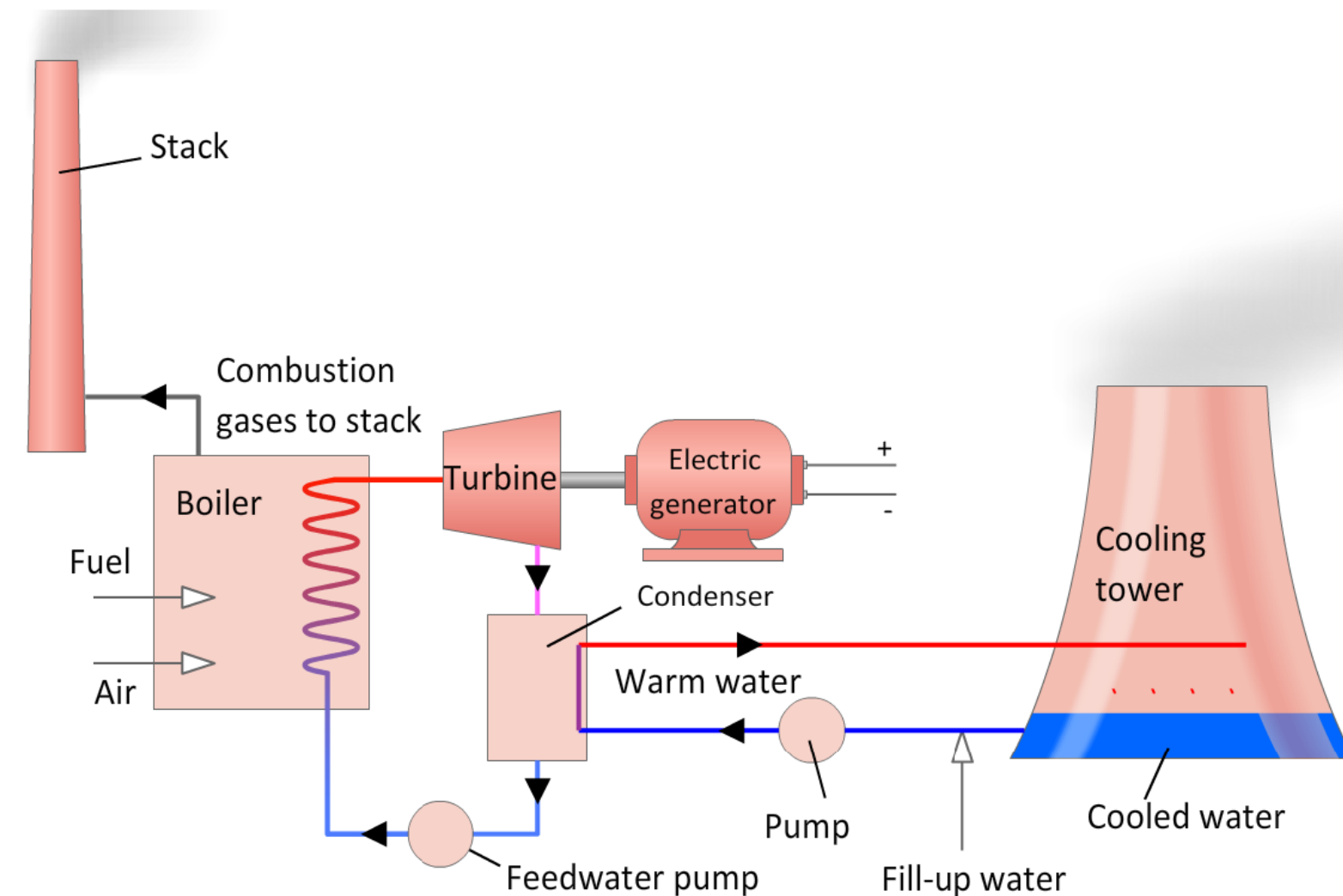
($q_{ab} = q_{dis}$)

Technical implementation

- ♦ Implementations of cycle processes
 - Steam power cycle process (Dt. Dampfkraftprozess)
 - Gas turbine cycle process

Steam Power Cycle

- ♦ Heterogeneous working material steam
- ♦ Boiler feed water pump (dt. Speisepumpe)
- ♦ Steam boiler (Dampferzeuger)
- ♦ Superheater (Überhitzer)
- ♦ Turbine
- ♦ Condenser (Kondensator)



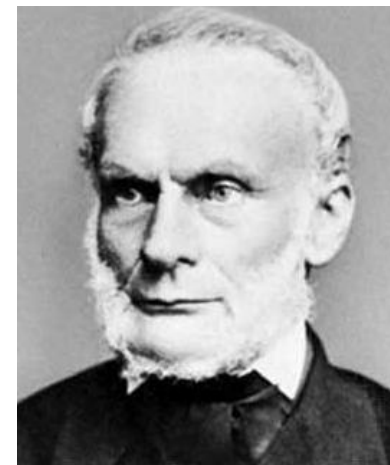
Rankine Cycle

- ♦ In German called “Clausius-Rankine” cycle

- ♦ William John Macquorn Rankine
 - Scottish mechanical engineer
 - 1820 – 1872

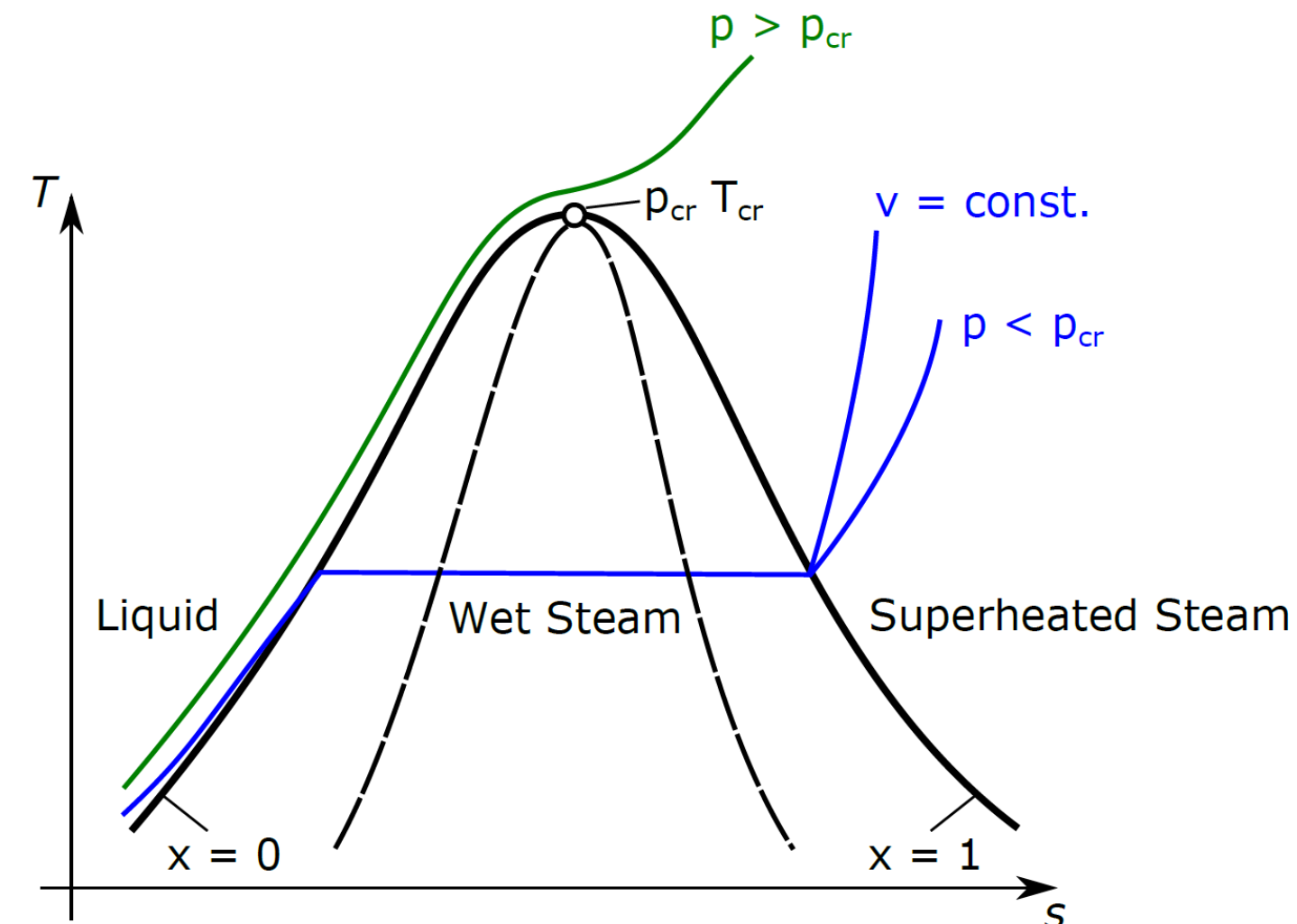


- ♦ Rudolf Julius Emanuel Clausius
 - German physician and mathematician
 - 1822 - 1888



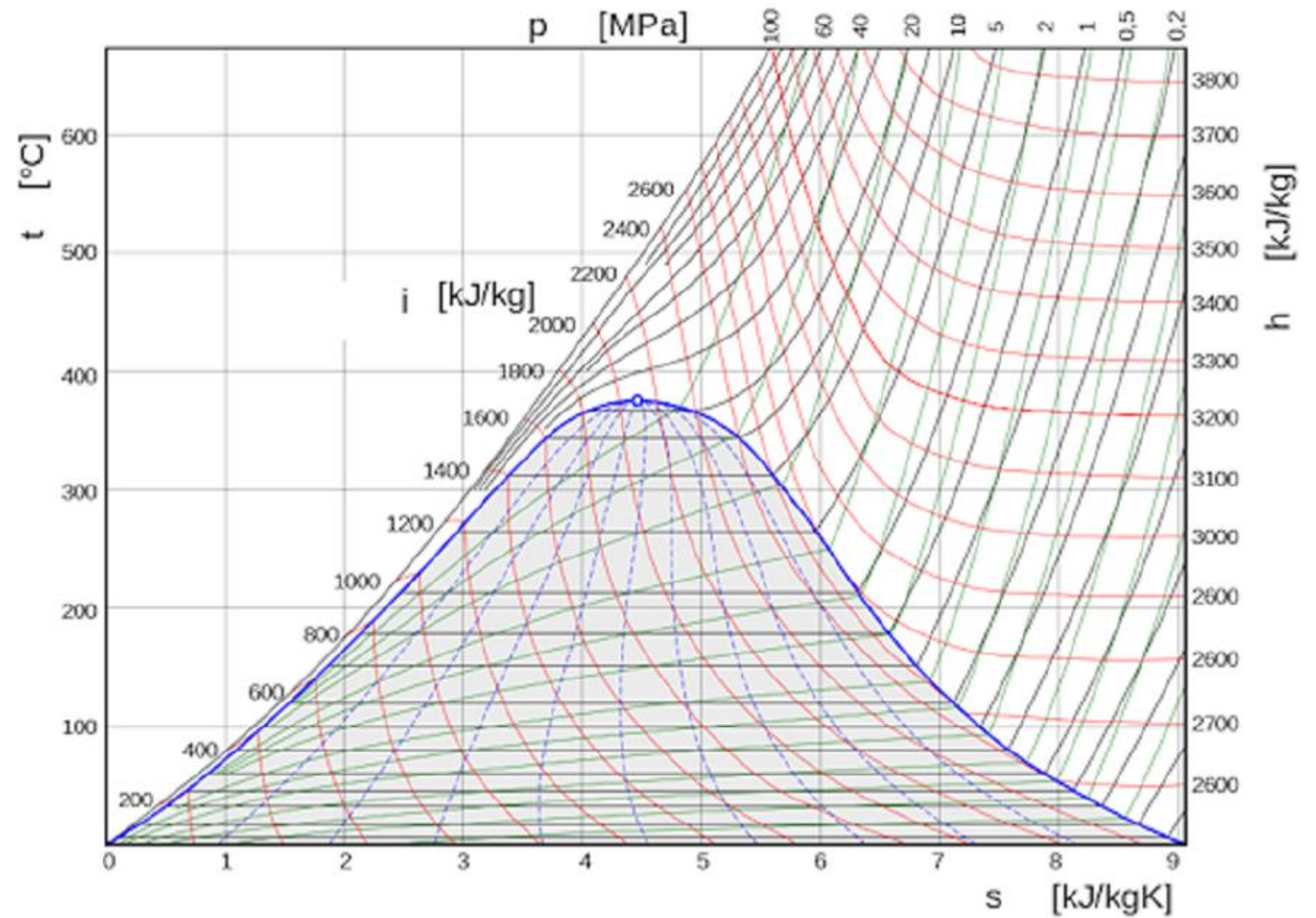
Properties of Water Steam

- ◆ Three aggregation states
 - Liquid (dt. Flüssigkeit)
 - Wet steam (dt. Naßdampf)
 - Superheated steam (dt. Dampf)
- ◆ Bell shaped boundary curve between them
- ◆ Lower boundary curve (Liquid-wet steam) dt. Siedelinie
- ◆ Upper boundary curve (Wet-steam Superheated steam) dt. Taulinie



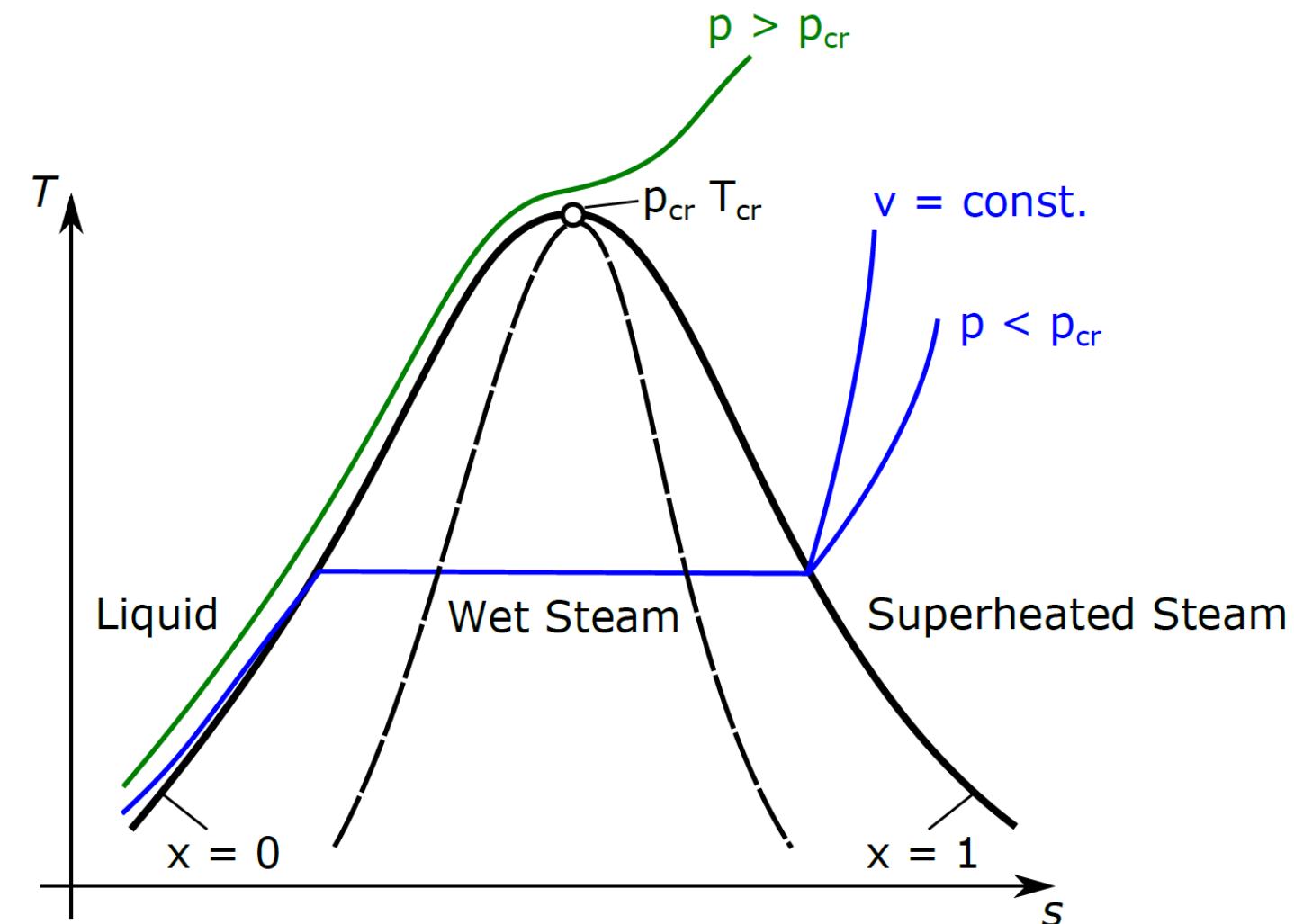
T-s Diagram

- ◆ T-s diagram is an extraction from a multi pane diagram



Properties of Water Steam

- ◆ As temperature increases the difference between aggregation states decreases
- ◆ On top of the boundary curve p_{critical} T_{critical}
 - No differences between aggregation states
 - Liquid instantly changes to vapour



Steam Tables (Dt. Wasserdampftafeln)

- ◆ Experimentally determined
- ◆ Values along lower boundary curve marked with ‘
- ◆ Values along upper boundary curve marked with “
- ◆ Used to calculate thermodynamical processes

States for liquid and superheated steam

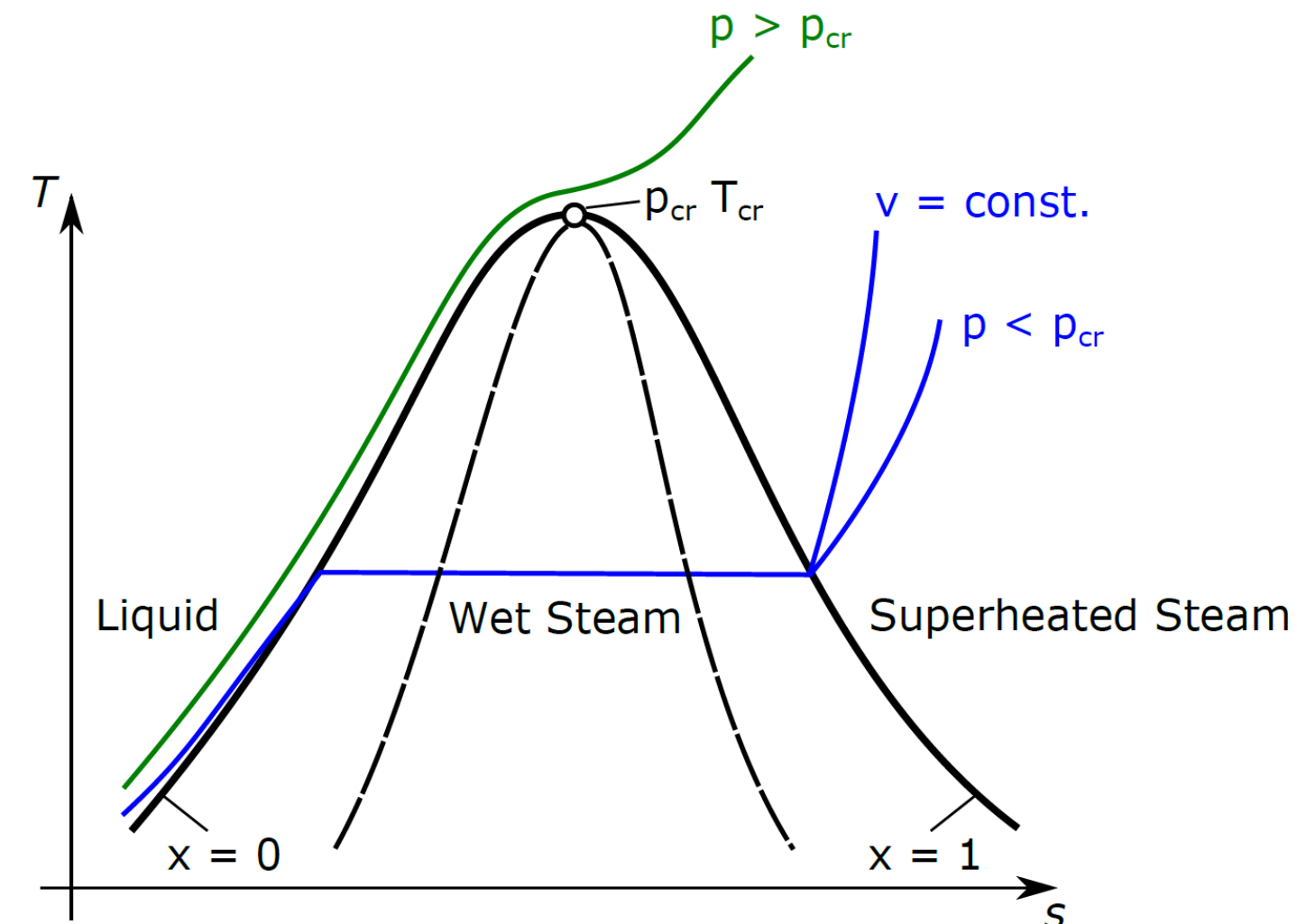
	230°C	240°C	250°C	260°C	310°C	530°C	540°C
5 bar							
v [m ³ /kg]	0,4549	0,4647	0,4744	0,4841	0,5321	0,7388	0,7481
h [kJ/kg]	2 919,1	2 940,1	2 961,1	2 981,9	3 085,4	3 548,4	3 570,1
s [kJ/kgK]	7,1903	7,2317	7,2721	7,3115	7,4971	8,1699	8,1967
40 bar							
v [m ³ /kg]	0,001207	0,001228	0,001251	0,05172	0,06044	0,09011	0,09135
h [kJ/kg]	990,5	1 037,7	1 085,8	2 835,6	2 990,2	3 513,1	3 535,8
s [kJ/kgK]	2,6077	2,7006	2,7934	6,1353	6,4130	7,1774	7,2055
150 bar							
v [m ³ /kg]	0,011924	0,001212	0,001232	0,001255	0,001421	0,02208	0,02250
h [kJ/kg]	993,1	1 039,2	1 086,2	1 133,9	1 394,5	3 394,3	3 421,4
s [kJ/kgK]	2,5867	2,6775	2,7681	2,8585	3,3250	6,4548	6,4885

States for wet steam

p	T	v'	v''	h'	h''	Δh_V	s'	s''
[bar]	[°C]	[m ³ /kg]	[m ³ /kg]	[kJ/kg]	[kJ/kg]	[kJ/kg]	[kJ/kgK]	[kJ/kgK]
0,10	45,833	0,0010102	14,67	191,83	2 584,8	2 392,9	0,6493	8,1511
0,13	51,062	0,0010126	11,47	213,70	2 594,0	2 380,3	0,7172	8,0592
5,0	151,84	0,0010928	0,3747	640,12	2 747,5	2 107,4	1,8604	6,8192
40	250,33	0,0012521	0,04975	1 087,4	2 800,3	1 712,9	2,7965	6,0685

Properties of Water Steam

- ♦ Wet Steam Region
 - Aggregation state of working medium is a two-phase mixture
 - Additional variable need to describe the dryness fraction x (dt. Dampfgehalt)
- ♦ Dryness fraction x
 - determine the ratio between saturated steam mass m_{vapour} to mass of mixture ($m_{\text{vapour}} + m_{\text{liquid}}$)



Dryness Fraction (Dt. Dampfgehalt)

- ♦ Dryness fraction x :

$$x = \frac{m_{vapour}}{m_{vapour} + m_{liquid}}$$

- ♦ To describe the content of the liquid in the wet steam mixture:

$$1 - x = \frac{m_{liquid}}{m_{vapour} + m_{liquid}}$$

- ♦ $1-x$ is called “degree of wetness”
- ♦ $x = 0$: Liquid water at the lower boundary curve
- ♦ $x = 1$: Saturated steam at the upper boundary curve

Calculations within wet steam region

- ♦ With dryness fraction x thermodynamic variables can be calculated

$$v = v' + x(v' - v'')$$

$$\rho = \rho' + x(\rho' - \rho'')$$

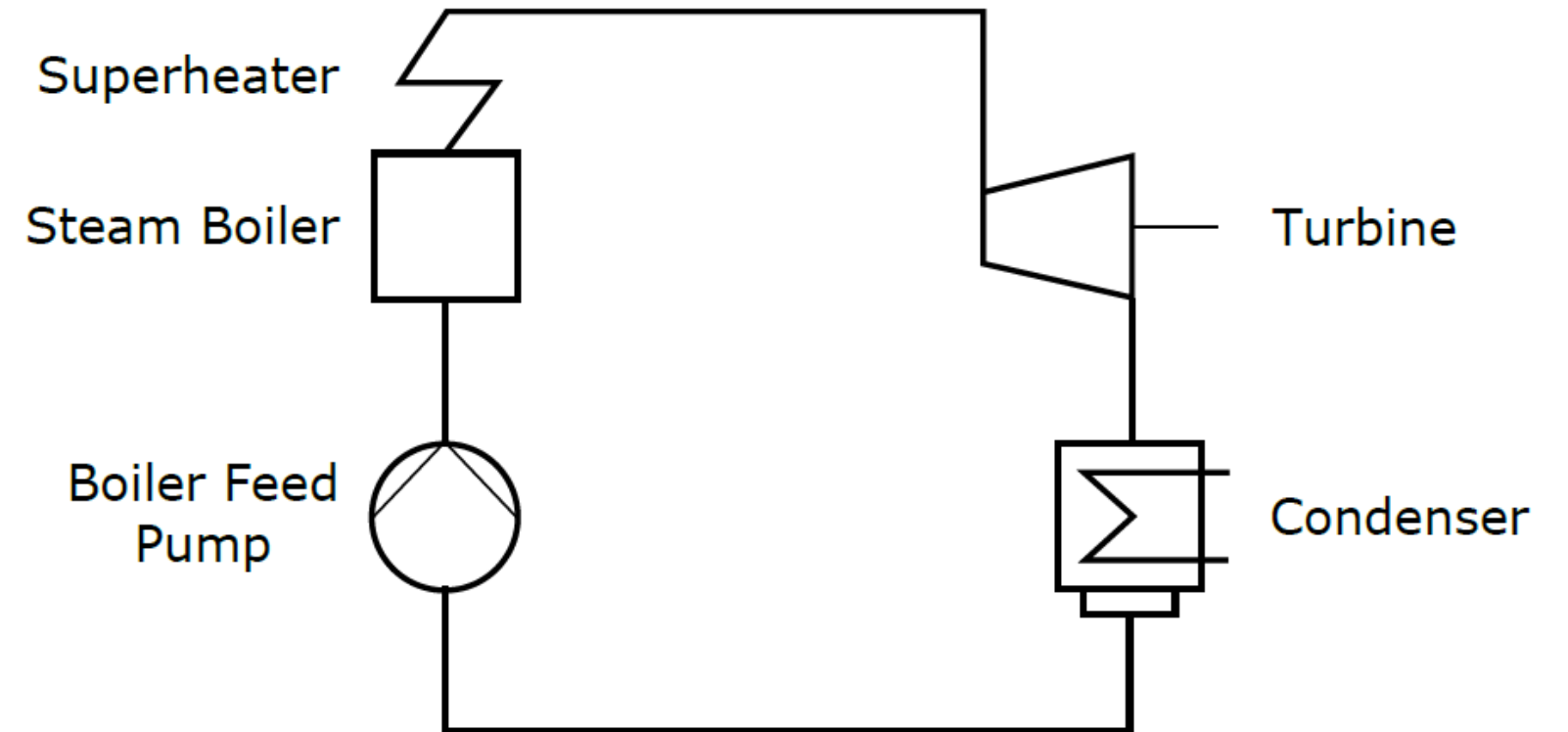
$$h = h' + x(h' - h'')$$

$$s = s' + x(s' - s'')$$

$$u = u' + x(u' - u'')$$

Ideal Rankine Process

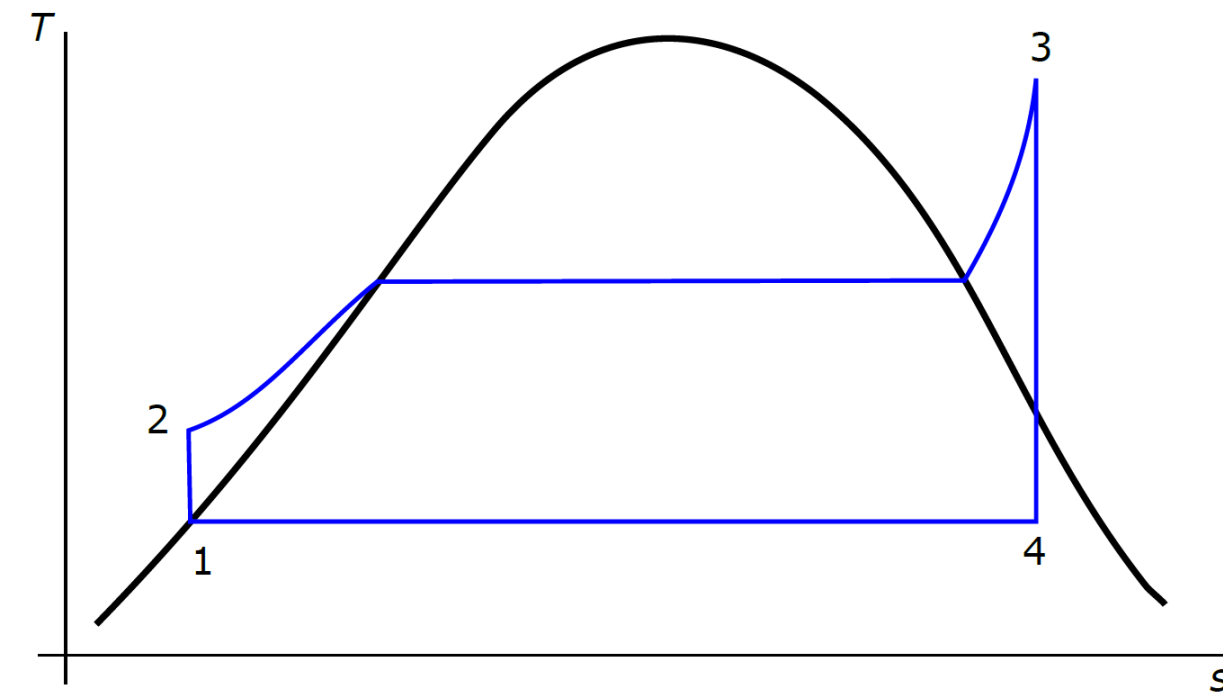
- ♦ Pump increases pressure
- ♦ High-pressure water flows into boiler
- ♦ Water heated up
 - until saturation temperature
 - vaporization begins
 - Additional heat until dryness fraction $x=1$
- ♦ Superheater heats up saturated steam again
- ♦ Superheated steam flows to turbine
- ♦ Within the turbine: Expands and high internal energy is converted to mechanical energy



- ♦ Turbine exit: Pressure and temperature are decreased
- ♦ Condenser
 - Remaining heat gets removed
 - Steam condenses
- ♦ Water flows to the intake of the pump

T-s Diagram

- ◆ 1-2: Isentropic ($s=\text{const.}$) compression of the water
- ◆ 2-3: Isobaric ($p=\text{const.}$) heating at constant pressure
 - Temperature rises until the saturation temperature for a given pressure
 - As water begins to boil, Temperature and pressure will be constant while dryness fraction rises until upper boundary curve
 - Steam gets superheated
- ◆ 3-4: Isentropic expansion
 - Temperature and pressure decreases
 - Steam becomes wet steam
- ◆ 4-1: Isobaric heat rejection and condensation



Carnot vs. Rankine

- ♦ Carnot cycle is important for historical and theoretical reasons
- ♦ In practice wet compression is difficult to archive
 - Carnot: pressure is raised through wet compression
 - Rankine: liquid is pump to the boiler with a pump, pump needs less work than a compressor
- ♦ Heat addition process
 - Carnot: Heat addition is isothermal
 - Rankine: Heat addition is isobaric, because isothermal heat addition is difficult to archive and if the expansion starts at the saturated vapor line, there will be erosion problems at the turbine

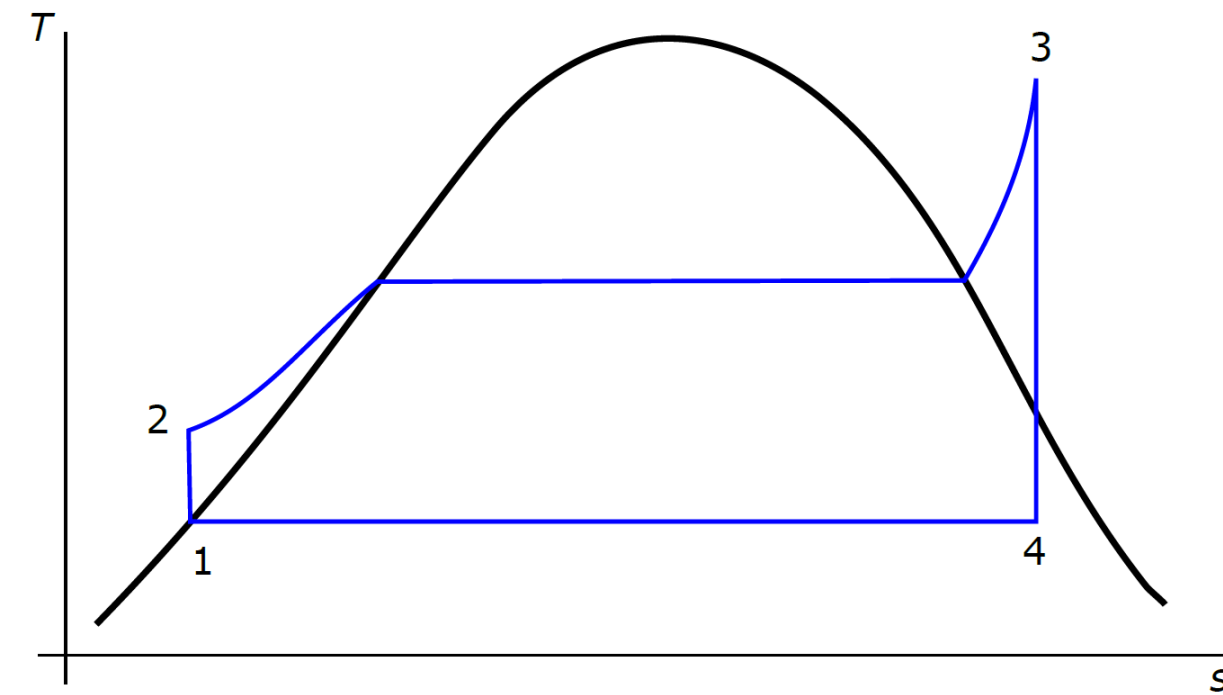
Heat Supply and Discharge

- ♦ Heat supply (2-3)
- ♦ Heat discharge (4-1)
- ♦ General equation

$$dh = du + pdv + vdp = dq + vdp$$

- ♦ Heat exchange happens during isobaric state
- ♦ Therefore $dp = 0$
- ♦ Term vdp becomes 0

$$dh = dq$$



Enthalpy differences

- ♦ ... continued
- ♦ Thus enthalpy differences dh are equal to heat differences dq
- ♦ Enthalpy h is directly converted to heat q
- ♦ Enthalpies h for different states:

$$dq_{sup} = dh_{Heating} = h_3 - h_2$$

$$dq_{dis} = dh_{Condenser} = h_4 - h_1$$

Work Supplied and Discharged

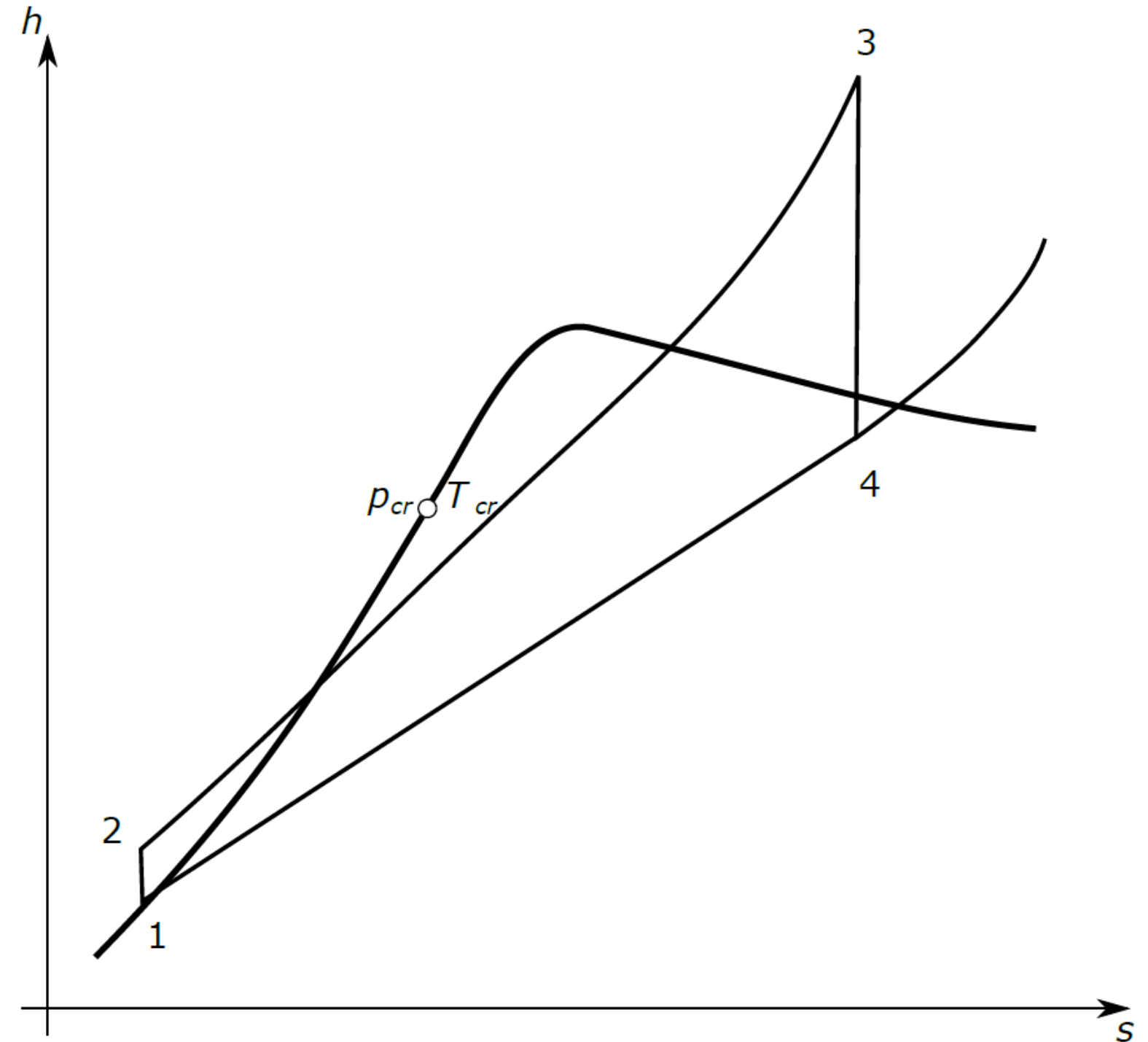
- ♦ Definition of a cycle process: applied heat and work extracted are equal.
- ♦ Thus the work exchange can be defined via enthalpy h :

$$dw_{sup} = dw_{Pump} = h_2 - h_1$$

$$dw_{dis} = dw_{Turbine} = h_3 - h_4$$

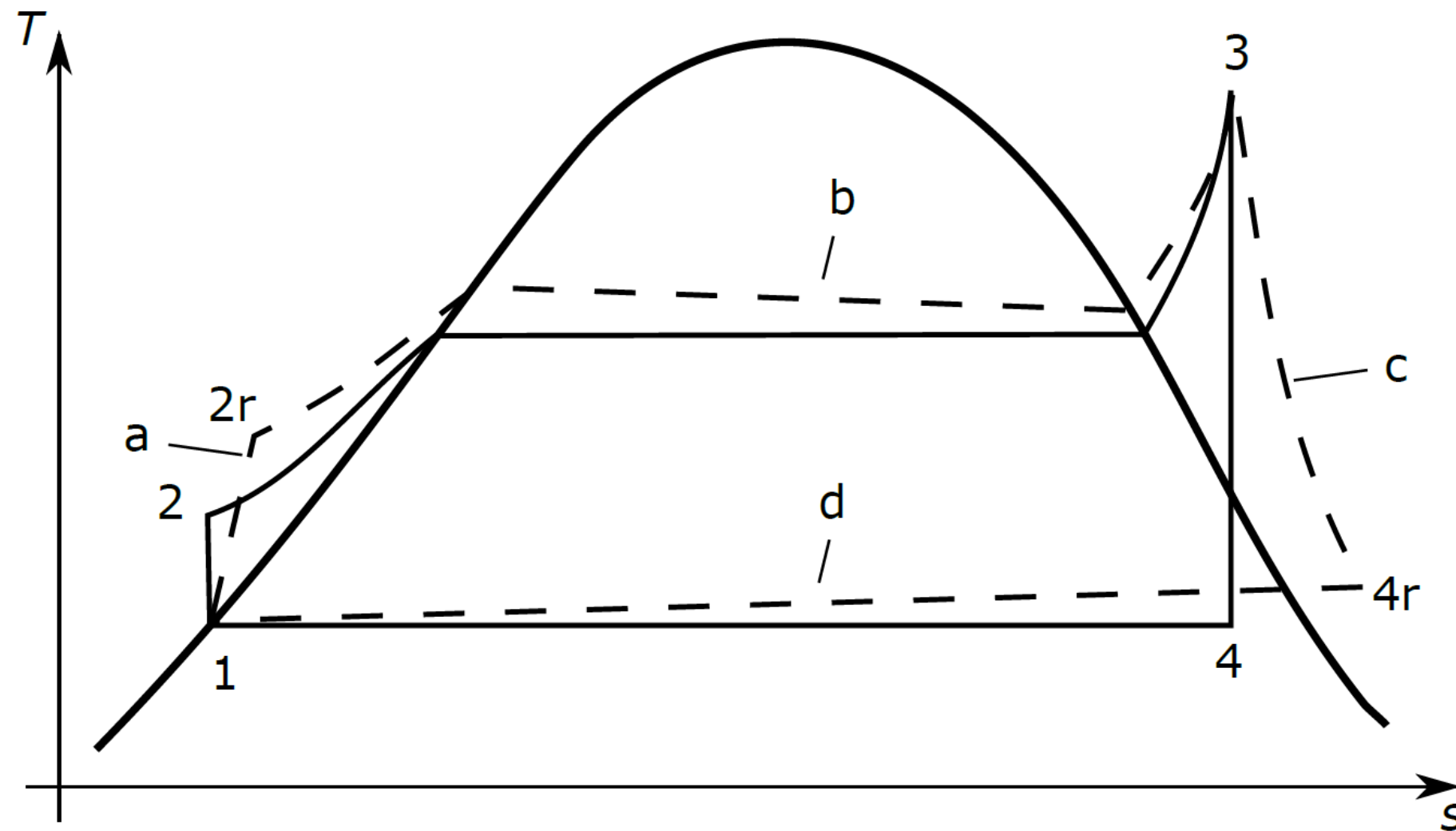
h-s Diagram of the Rankine cycle

- ♦ The h-s diagram can be used easily get the enthalpies h for different states



The Real Rankine process

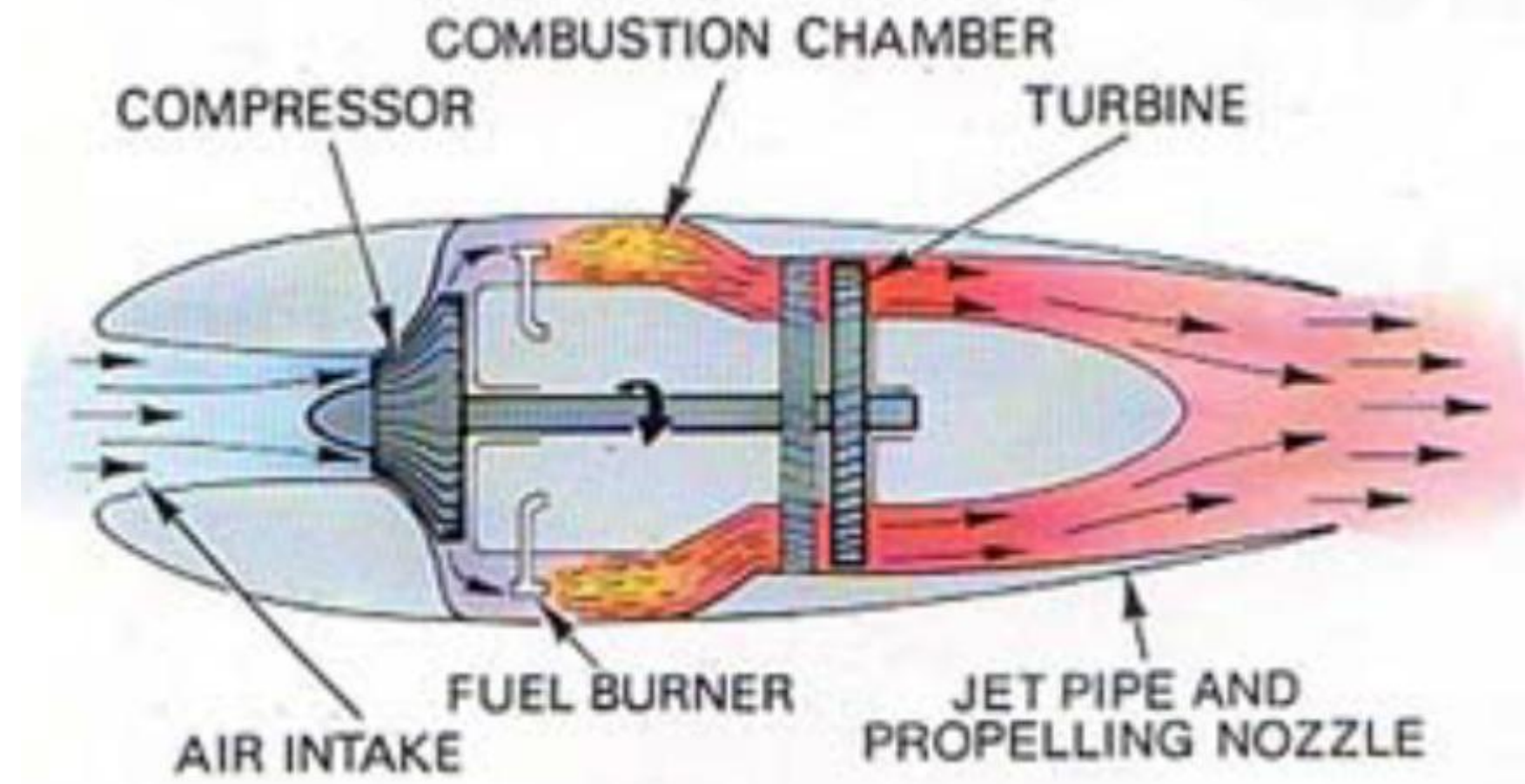
- ♦ Ideal process
 - working medium is reversible
- ♦ Real process
 - Irreversible compression in boiler feed pump
 - Pressure decrease during heating
 - Irreversible expansion in the turbine
 - Pressure decrease in the condenser



Solid line – ideal
Dashed line - real

Gas Turbine Cycle

- ◆ Used in
 - Aircraft propulsion
 - electric power generation
 - Reduction gear needed
 - Turbine speed 12 000 rpm
 - Sync. generator speed
 - 1000 (50 Hz grid)
 - 1200 rpm (60 Hz grid)
- ◆ Homogeneous working medium
- ◆ Cycle process: Brayton cycle



Feature of a Gas Turbine

- ♦ High thermal efficiencies (up to 44%)
- ♦ Suitable for combined cycles (heating up steam as well)
- ♦ High power to weight ratio (aircraft usage)
- ♦ Fast start up time
 - Around 2 minutes
 - Steam system needs up to 4 hours

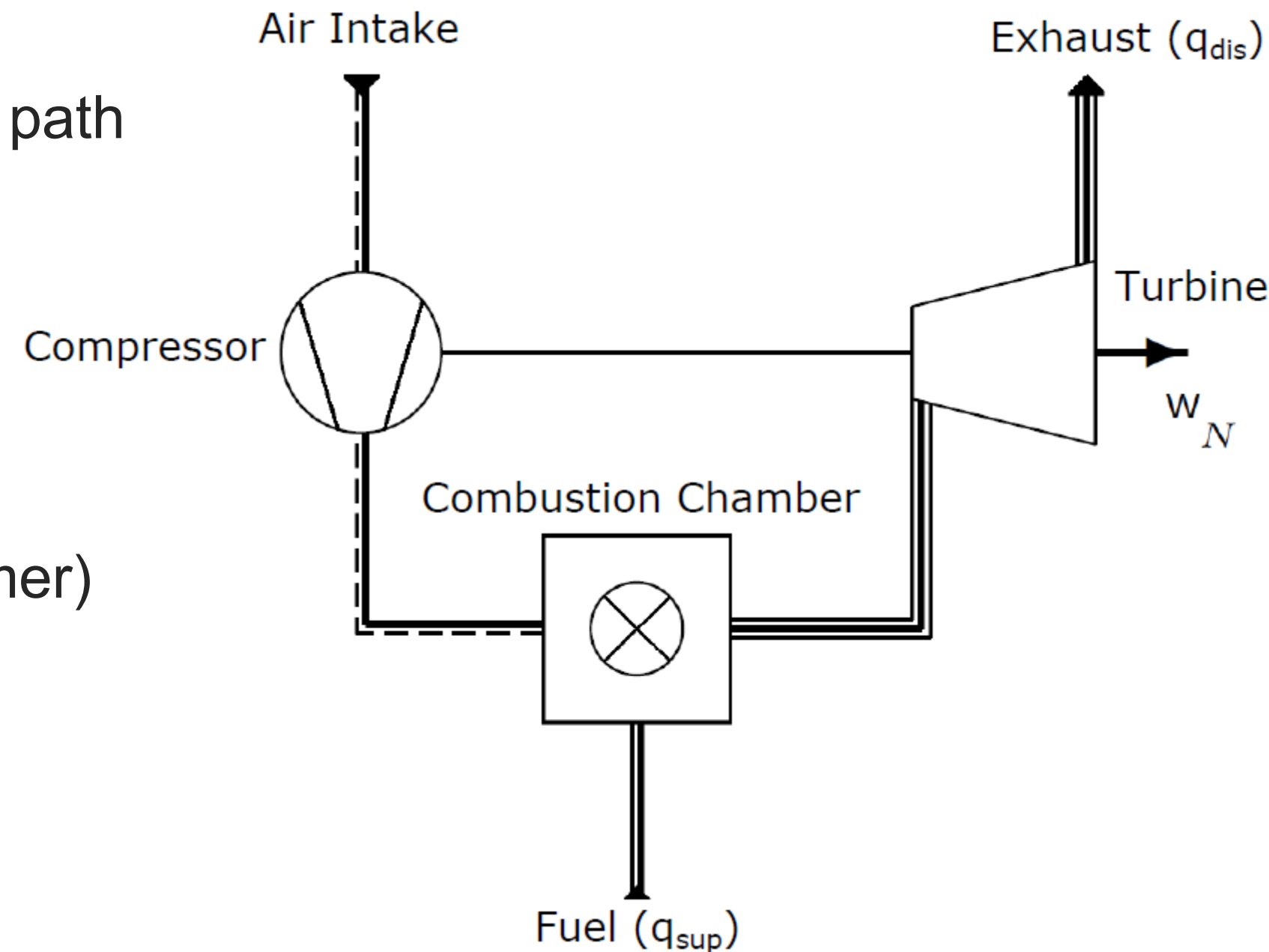
Bryton Cycle

- ♦ In German called “Joule” cycle
- ♦ James Prescott Joule
 - Physicist and brewer from UK
 - 1818 – 1889
- ♦ George Baily Brayton
 - Mechanical engineer from US
 - 1830 - 1892



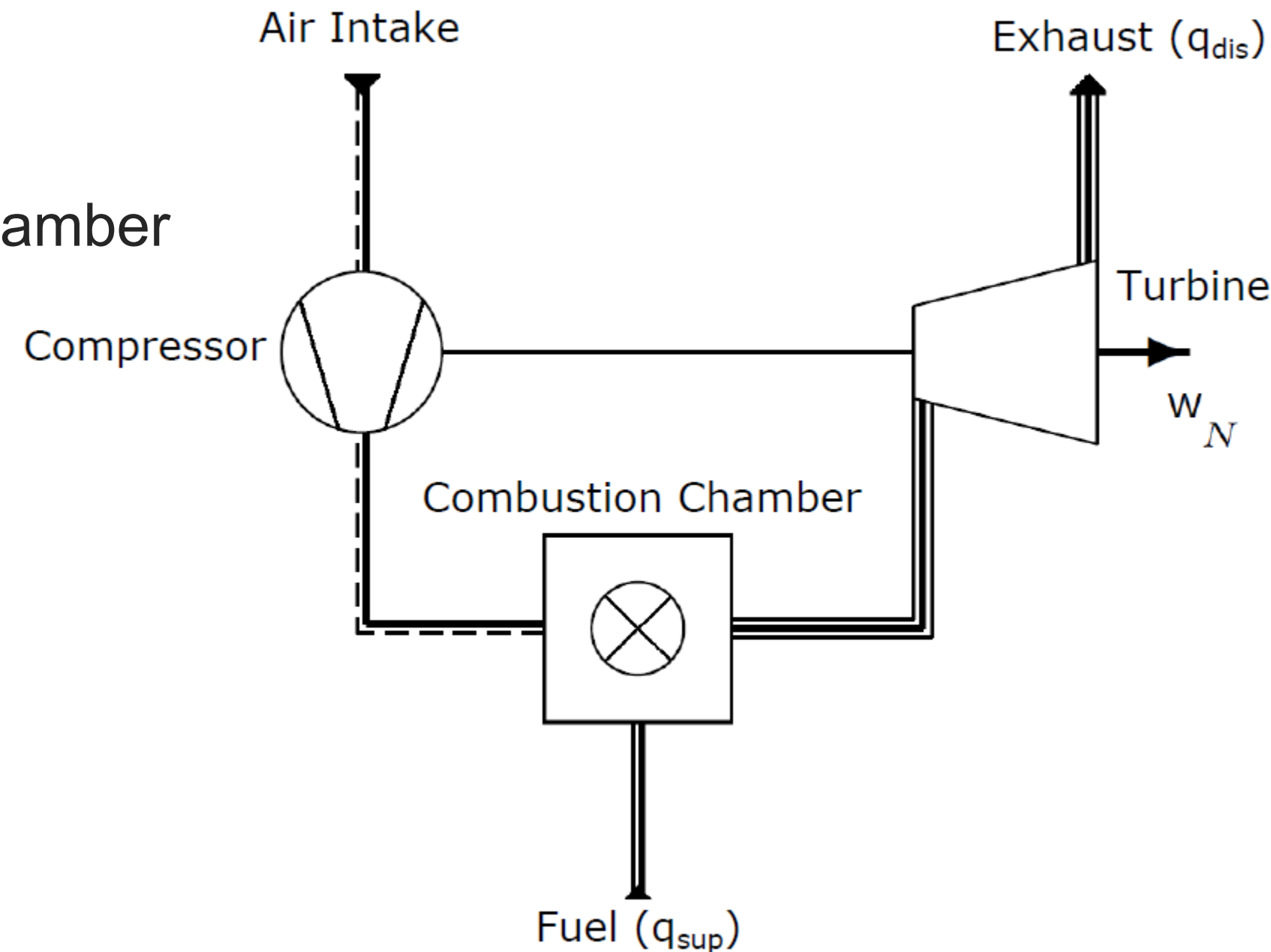
Ideal Brayton Cycle

- ♦ Ideal cycle process
- ♦ Working medium moves along a closed path
- ♦ Used components
 - Air intake (dt. Frischluft-Einlass)
 - Compressor (dt. Verdichter)
 - Combustion Chamber (dt. Brennkammer)
 - Fuel (dt. Brennstoff)
 - Turbine
 - Exhaust (dt. Abgas)



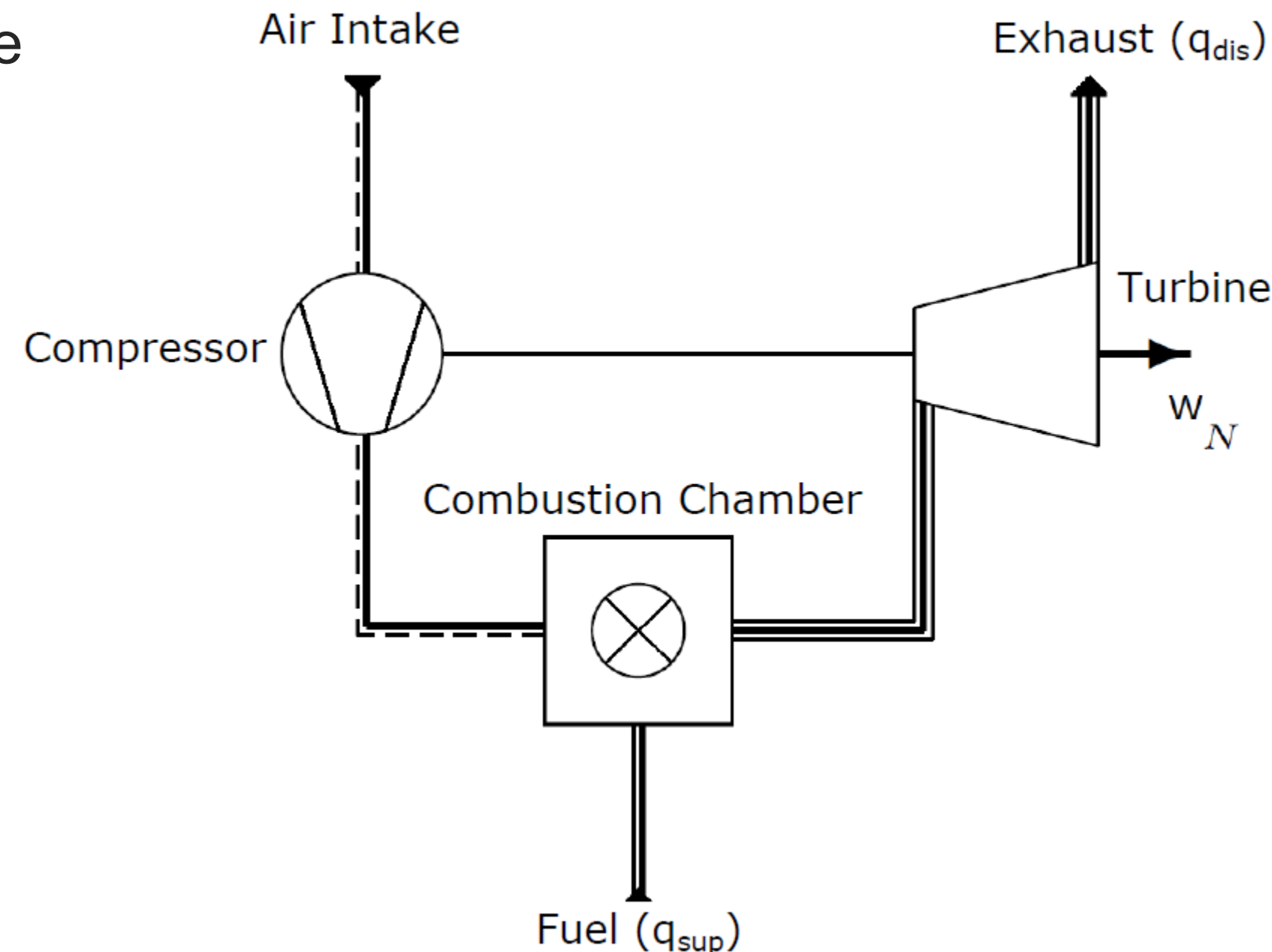
Ideal Brayton Cycle

- ♦ Ambient air drawn into the machine
- ♦ Air is compressed
- ♦ Feed at high pressure to combustion chamber
- ♦ Fuel injected – air is heated up
- ♦ Hot air expands in the turbine
- ♦ Leaves the gas turbine via exhaust



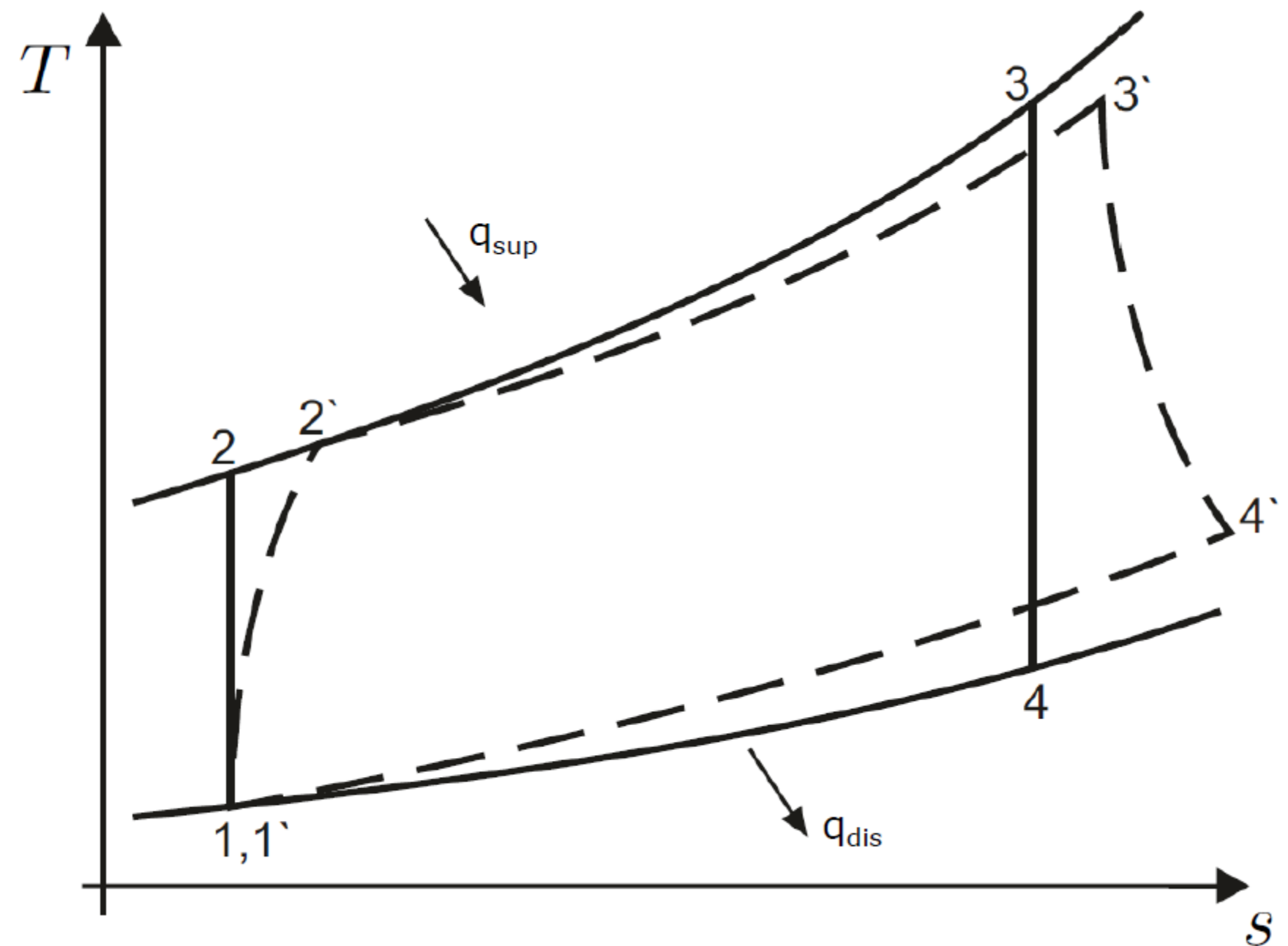
Ideal Brayton Cycle

- ♦ Air and fuel are added
 - Classified as open system
- ♦ Compressor and turbine mounted on the **same shaft**
- ♦ Turbine speeds up – compressor is forced to speed up and increases burning rate by sending more high pressure and hot gases to the combustion chamber – Turbine speeds up again
- ♦ Fuel control needed to prevent uncontrolled runaway of the speed



T-s Diagram

- ♦ 1-2 Isentropic ($s=\text{const.}$) compression of the air by the compressor
- ♦ 2-3 Isobaric ($p=\text{const.}$) heating at constant pressure in the combustion chamber
- ♦ 3-4 Isentropic expansion in the turbine
- ♦ 4-1 Isobaric heat rejection in the exhaust



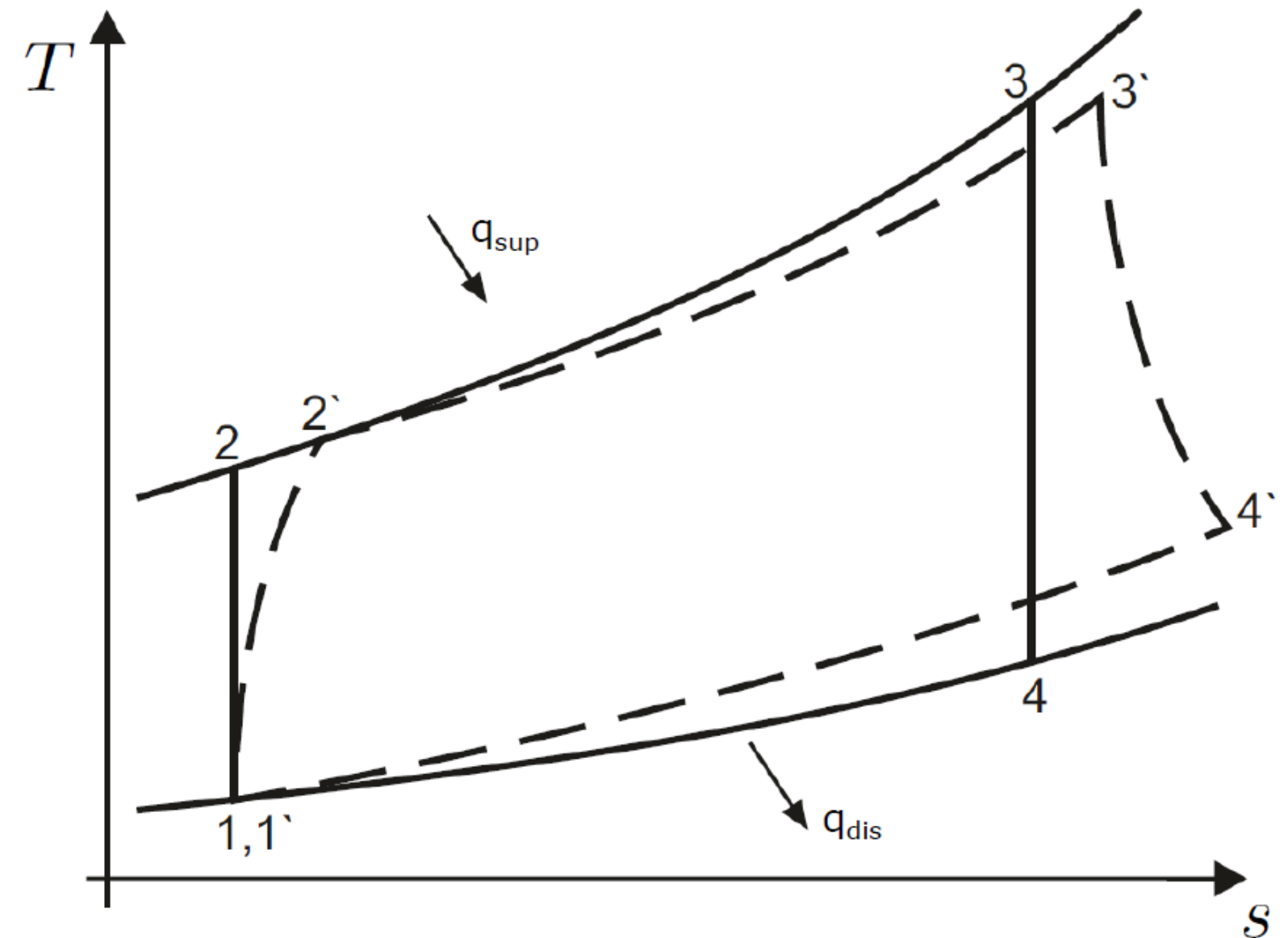
Solid line – ideal
Dashed line - real

Heat Supply and Discharge

- ◆ Heat q_{sup} gets supplied between 2-3
- ◆ Heat q_{dis} gets discharged between 4-1

$$q_{\text{sup}} = h_3 - h_2 = c_p(T_3 - T_2)$$

$$q_{\text{dis}} = h_1 - h_4 = c_p(T_1 - T_4)$$



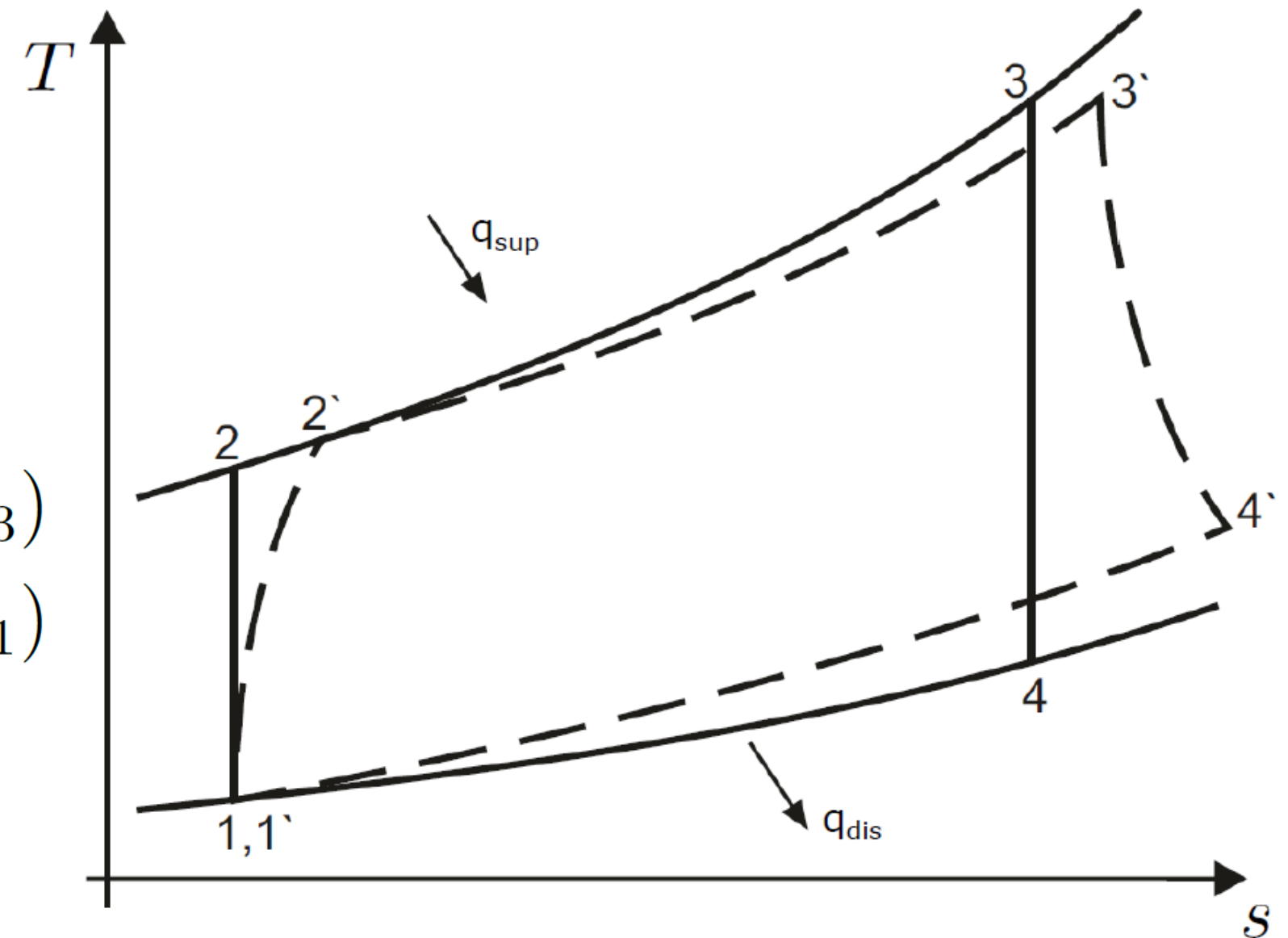
Solid line – ideal
Dashed line - real

Work Supply and Discharge

- ♦ Compression work $w_{\text{Compressor}}$ gets applied between 1-2
- ♦ Useful work w_{Turbine} gets withdrawn between 3-4

$$w_{\text{Turbine}} = h_4 - h_3 = c_p(T_4 - T_3)$$

$$w_{\text{Compressor}} = h_2 - h_1 = c_p(T_2 - T_1)$$



Solid line – ideal
Dashed line - real

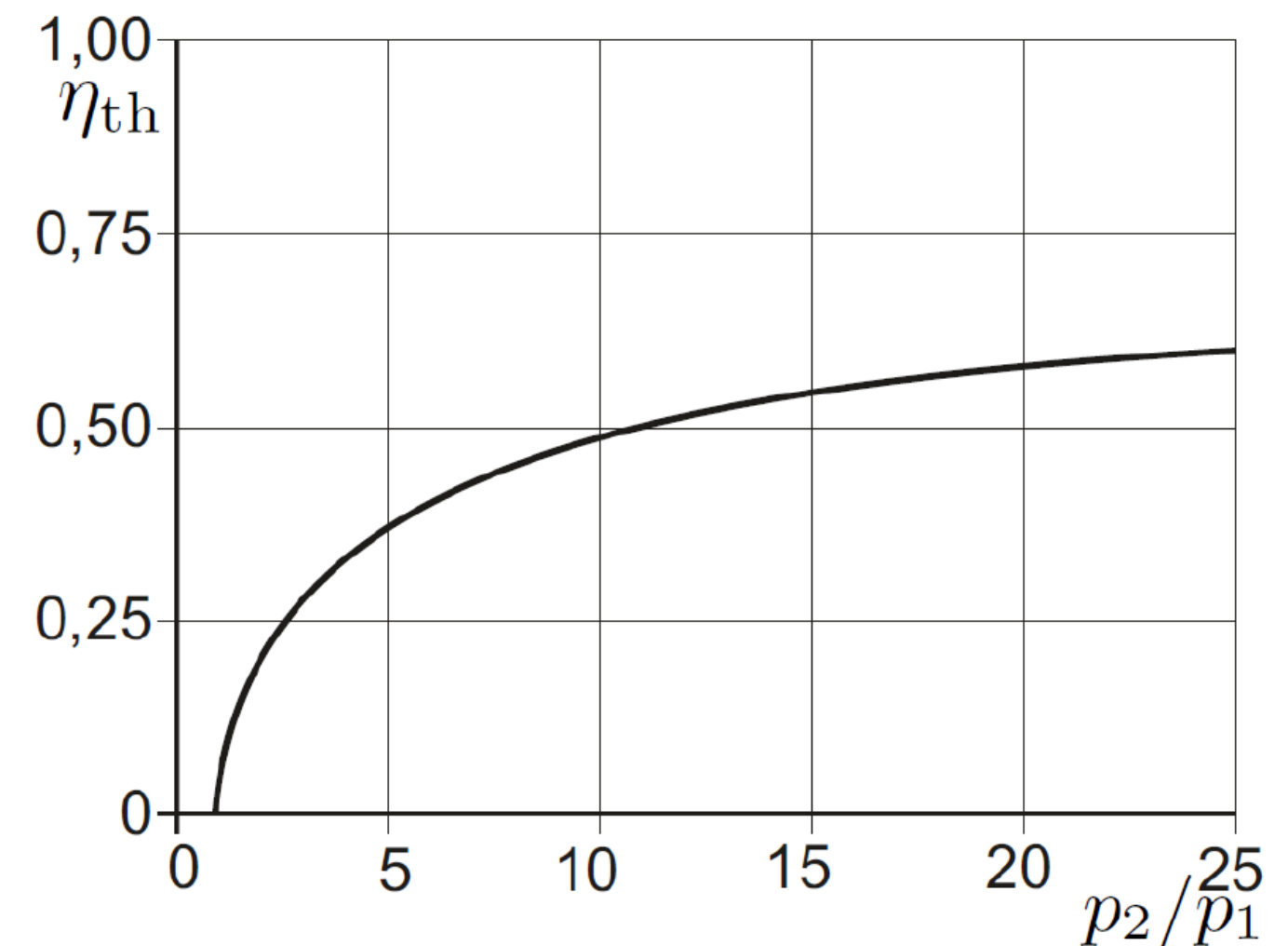
Useful Work

- ♦ Useful work which can be withdrawn from the cycle:

$$w_U = w_{Turbine} + w_{Compressor} = q_{sup} - q_{dis}$$

Efficiency factor

- ◆ Efficiency factor is based on pressure ratio
- ◆ Carnot cycle maximizes temperature differences
- ◆ Brayton cycle maximizes pressure differences



Heat influence

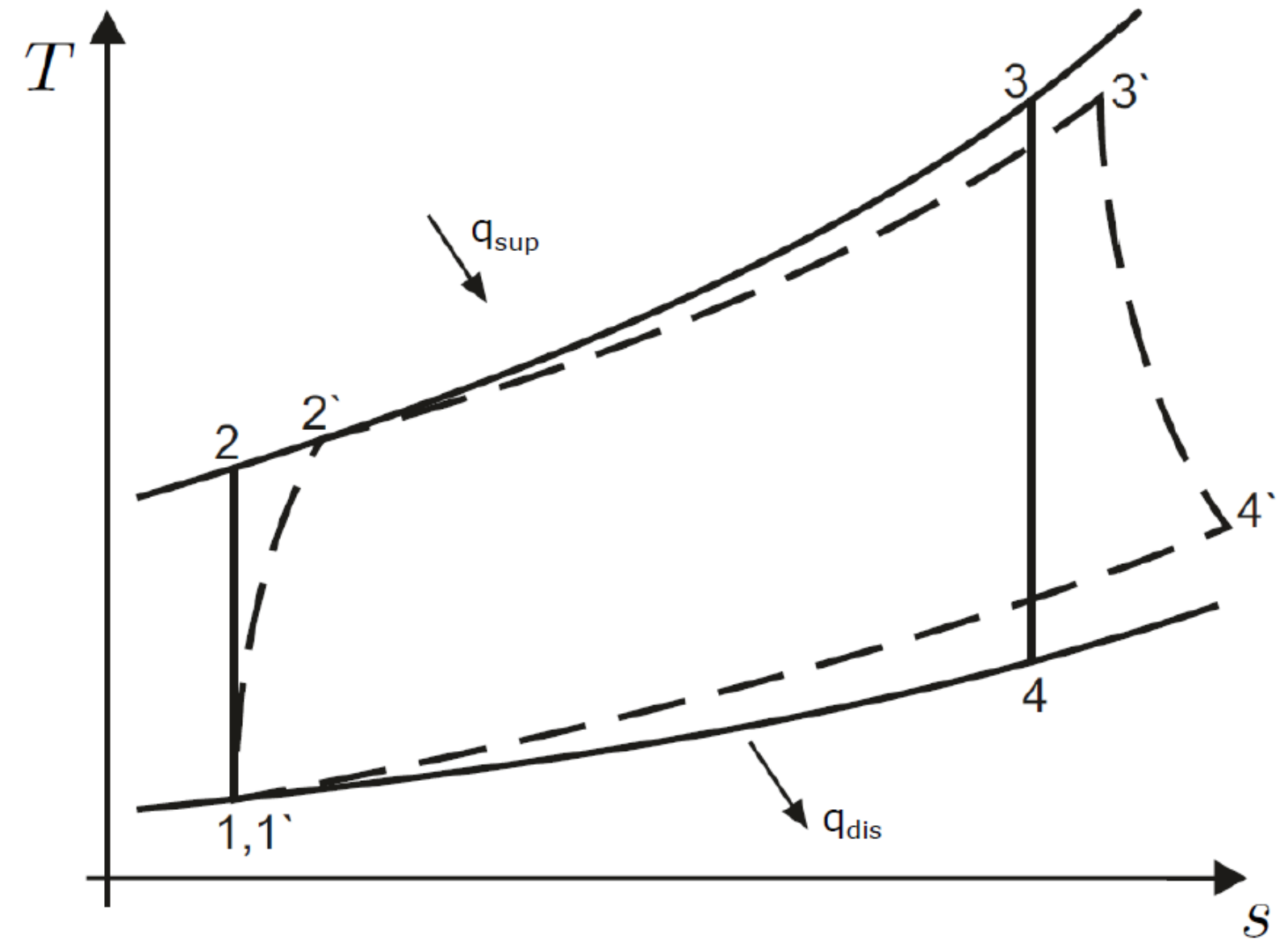
- ♦ Amount of supplied heat in the combustion chamber has no impact on efficiency factor
- ♦ However, is important for useful work output:

$$|w_U| = |w_{Turbine}| - |w_{Compressor}| = \eta_{th} q_{sup}$$

- ♦ To get enough work output, the machine needs to be sufficiently fueled

The real Brayton process

- ♦ Implementation of ideal process has technical limitations
- ♦ Irreversibilities within the turbine (like for the steam process)
- ♦ Compressor losses
- ♦ Pressure losses in the combustion chamber



Solid line – ideal
Dashed line - real



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